

1 TITLE:	Thermodynamics & Phase Equilibria in Ceramic Systems
1.2 *COURSE NUMBER (if known)::	DC.MCR 202.15
1.3 CREDITS::	11(L-T-P : 3-1-0)
1.4 SEMESTER-OFFERED::	III semester

2. OBJECTIVE::

This course has been designed to expose the students about the fundamental and applied knowledge of thermodynamics and phase equilibria in ceramic systems .

3. COURSE TOPICS::

1. Review of Fundamentals: Introduction, definition of terms, first law of thermodynamics: Heat and work, internal energy, isometric process, isobaric process, isothermal process and enthalpy, heat capacity. Second law of thermodynamics: spontaneous process, entropy and irreversibility, entropy and reversible heat, reversible isothermal compression, adiabatic expansion of ideal gases, second law of thermodynamics, maximum work, criterion of equilibrium, combined statement of first and second laws. third law of thermodynamics.

2. Statistical interpretation of entropy: Entropy and disorder, microstate, most probable state and equilibrium, Boltzmann equation, thermal entropy and configurational entropy.

3. Thermodynamics behaviour of solutions: Raoult's law and Henry's law, The thermodynamics activity, Gibbs-Duhem equation, Gibbs free energy of formation of a solution, properties of ideal solutions, non ideal solutions, Gibbs-Duhem equation and activity relationship, regular solutions. A statistical model for solution, sub regular solutions.

4. Phase equilibrium in a one component system: Variation of Gibbs free energy with temperature and pressure, equilibrium between different phases- solid liquid equilibrium, Clapeyron equation, Clausius-Clapeyron equation. Graphical representation of equilibrium in one component system.

5. Two component system: Gibbs free energy-composition diagrams and phase equilibrium. Gibbs free energy and thermodynamics activity, Gibbs free energy of formation of regular solutions, criteria for phase stability, continuous solid solution, eutectic reaction, liquid phase separation, peritectic reactions, compound formation; congruently and incongruently melting compound.

6. Electrochemistry: chemical reactions and electrochemical reactions, chemical and electro- chemical driving forces. Electrochemical cell- EMF, different types of electro chemical cells, stabilized zirconia as solid electrolyte, oxygen sensor, solid oxide fuel cells.

4. READINGS

4.1 TEXTBOOK: 1. Introduction to thermodynamics of materials: D.R. Gaskell

4.2 *REFERENCE BOOKS:: 1) Physical Chemistry of metals: L.S. Darken and R. W. Gurry